

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C02: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A sDry

Run Evaluation

Code of Conduct:

I G . Hema chowdary(192473014) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

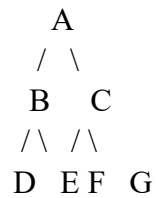
G Hema chowdary

Annexure A

Dry Run Evaluation

1. Breadth-First Search (BFS)

Graph :



BFS Dry Run :

Iteration	Queue Before	Visited Node	Queue After
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

Answers

1. BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

2. Queue after each iteration

- B, C
- C, D, E
- D, E, F, G
- E, F, G
- F, G
- G
- Empty

2. Depth-First Search (DFS)

DFS Traversal

Following the left subtree first:

Step	Stack (Conceptually)	Visited
1	A	A
2	B	A, B
3	D	A, B, D
4	Backtrack	A, B, D
5	E	A, B, D, E
6	Backtrack to A	A, B, D, E
7	C	A, B, D, E, C
8	F	A, B, D, E, C, F
9	Backtrack	A, B, D, E, C, F
10	G	A, B, D, E, C, F, G

DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

3. 8 Puzzle using Breadth-First Search

Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

BFS Dry Run :

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	1	4 successor states
2	First generated state	4 states	New successors added
3	Second generated state	Remaining states	Expanded further
4	Third generated state	Updated queue	Updated queue
5	Fourth generated state	Updated queue	Updated queue

Answers

1. Which node is expanded in each iteration?

- Iteration 1 – Initial state
- Iteration 2 – First child generated
- Iteration 3 – Second child generated
- Iteration 4 – Third child generated
- Iteration 5 – Fourth child generated

2. How many states are generated in total?

The center blank tile has 4 possible moves (Up, Down, Left, Right) from the initial state, so 4 successor states are generated initially. As BFS continues, additional unique states are generated until the goal is found.

3. Complete Solution Path :

Initial State

1 3 6
5 0 2
4 7 8

↓

Intermediate State(s)

↓

Goal State

1 2 3

4 5 6

7 8 0

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle?

Breadth-First Search explores all states level by level. Since each move has the same cost, BFS reaches the goal using the minimum number of moves before exploring deeper levels. Therefore, it always finds the shortest path (optimal solution) in an unweighted state space.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.